Paper II: formalization and reproducibility

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Paper II · Formal companion v1.3
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This document accompanies the mathematical article. It preserves the formal declarations, scope qualifications and computational evidence previously included in the article. The source freeze and its recorded verification are unchanged. No Lean/Lake build was run for this edition. Historical section and table numbers in the records below refer to the preceding article; they are retained as locators, not renumbered as new verification results. Original references [1], [2], etc. are listed at the end.

Frozen sources and historical evidence / Fuentes congeladas y evidencias históricas · Previous article / Artículo anterior

Corollary 1.2′ (sharpened asymptotic)

Formalization status. The delivered Paper II Lean v1.2 freeze uses Lean 4 / Mathlib v4.28.0 [5,6]. Its recorded main build completed 8,061 jobs, and the explicit extremizer/copy-defect supplement completed 8,032 jobs. The main declaration is named PaperII.theorem_1_2; it corresponds to Theorem 1.1 of this manuscript and includes the accompanying attainment statement. The same freeze includes the complete-split argmax, level sets, copy defects, and PaperII.AsymptoticCorollaries. Its theorem-level axiom gate reports no project-level mathematical axiom. The formal development derives the chordal structure used in the proof from the standard predicate IsChordal; no clique-tree or chordal-structure axiom is added. Section 8 records the exact freeze perimeter.

8. Reproducibility, formal verification, and scope

The proof in the paper is analytic. Computational experiments were used during development to search for counterexamples and to catch transcription errors, but no computation is used as a premise in the proof.

The accompanying calculation ledger records the proof at a lower level of granularity. It lists the definitions, the copied-cover construction in Lemma 3.1, the two cancellation identities, the discrete-convexity lift to clone classes, the clone-progress argument, the simpliciality requirement in the chordal reduction, the terminal complete-split characterization, the two-variable program on $S_{p,q}$, the degenerate case $S_{2,0}$, the formulas for $\Phi_\tau(S_{p,q})$, and the final integer maximization. The theorem, the exact bound, and the assembly have received internal adversarial audit. This records the author's validation process and is not a claim of external peer review.

Formal verification in Lean

The consolidated snapshot at PAPER_II/05_formalization/lean_v1.2_freeze uses Lean 4 / Mathlib v4.28.0 [5,6]. The recorded main build completed 8,061 jobs with zero errors; an explicit supplement covering PaperII.Extremizer and PaperII.CopyDefect completed 8,032 jobs with zero errors. The corresponding main declaration is

```
PaperII.theorem_1_2
```

in repository [7]. The formal theorem includes both sides of the result: the upper bound over all chordal graphs on n vertices and the existence of a complete-split graph attaining the value.

The formalization uses the standard chordal predicate IsChordal. In the development, this predicate says that every cycle of length at least four has a chord. The deficit functional is formalized as

```
phiTau G = (|E(G)| : ℝ) - 2 · tau3star G,
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where tau3star is the fractional triangle-cover number of the finite cover LP. The complete-split family is formalized as completeSplit\p\q, and the extremal value is the integer $\lfloor (2n+1)^2/24 \rfloor$.

No chordal-structure axiom is added. The formalization derives from IsChordal the hereditary property for induced subgraphs, preservation under adding a vertex with clique neighborhood, and the existence of simplicial vertices. The written proof of Lemma 5.1 uses clique trees as standard textbook infrastructure; the Lean proof uses a clique-tree-free terminal-characterization argument. Thus the formal proof does not assume a clique-tree or running-intersection axiom.

The checked development covers the copied-cover inequality, clone-class discrete convexity, clone-class progress, chordality-preserving copying, the terminal complete-split characterization, orbit averaging on $S_{p,q}$, the terminal linear program, and the exact integer maximization. The freeze log records 49 linter warnings, principally unused section variables and unused simp arguments; these do not affect kernel elaboration. The recorded axiom report is

```
#print axioms PaperII.theorem_1_2

[propext, Classical.choice, Quot.sound]
```

These are Lean's standard classical axioms. The report contains no sorryAx and no project-specific axiom. The frozen source archive is PAPER_II_lean_v1.2_freeze.zip, with SHA-256 ee2d05cc40d943ca92f8f7bf3e5dd83c2692518ddea5e2ca4f7686ccb1ac3895. The main and supplement logs are stored as gate_logs/BUILD_LOG.txt and gate_logs/BUILD_SUPPLEMENT_EXTREMIZER_COPYDEFECT.txt within lean_v1.2_freeze. The chordal-graph contribution modules shipped in the same archive are candidates for upstreaming to Mathlib and are not additional proof assumptions of this paper. The frozen archive passed independent clean-room reproduction; the external audit recorded a successful 8,063-job build and the expected 16 axiom surfaces.

Table 4. Frozen formalization perimeter (Lean v1.2). The delivered freeze contains the main theorem and the manuscript declarations listed below; independent clean-room reproduction returned PASS.

Node	Statement	Axiom footprint
PaperII.theorem_1_2	$\max_{\text{chordal}, \lvert V\rvert=n} \Phi_\tau = \lfloor (2n+1)^2/24 \rfloor$, attained by a complete-split graph	propext, Classical.choice, Quot.sound
Fsat_argmax_unique, Fsat_argmax_tie	uniqueness of the maximizing integer p within the complete-split family, and the two-value tie when $n \equiv 1 \pmod 3$	propext, Classical.choice, Quot.sound
level_set_iff (Cor. 7.2)	exact level-set equivalence from the continuous maximum	propext, Classical.choice, Quot.sound
copyDefect_nonneg, copyGamma_ge_half_copyDefect	copy defect $\Delta_{\Phi_\tau} \geq 0$ and $\Gamma_{\Phi_\tau} \geq \Delta_{\Phi_\tau}/2$	propext, Classical.choice, Quot.sound

Table 5. Arithmetic surface included in the Lean v1.2 freeze. These declarations are consequences of the frozen extremal value and are included in the delivered archive.

Node	Statement	Axiom footprint
phiTau_max_sandwich (Cor. 1.2′)	for every $n \in \mathbb{Z}$, $4n^2+4n-23 < 24\lfloor (2n+1)^2/24 \rfloor \leq 4n^2+4n+1$	propext, Classical.choice, Quot.sound
odd_sq_emod_24, phiTau_max_closed (Cor. 7.3)	$(2n+1)^2 \equiv 1$ or $9 \pmod{24}$, and the resulting residue-class formula for the extremum	propext, Classical.choice, Quot.sound
phiTau_max_le_paperI_bound	$\lfloor (2n+1)^2/24 \rfloor \leq n^2/6 + n/2$ for $n \geq 1$ (comparison with Paper I)	propext, Quot.sound

The four declarations in Table 5 are collected in PaperII/AsymptoticCorollaries.lean, which builds against the same toolchain (Lean 4 / Mathlib v4.28.0) and carries only Lean's standard classical axioms, with no sorryAx and no project-specific axiom. The per-declaration #print \axioms reports are archived in PaperII/AxiomCheckCorollaries.lean. The module adds no hypothesis to and does not modify the frozen main development; each declaration is derived from the closed value by ring, a finite residue split, and omega.

Independent computational verification

In addition to the analytic proof and the Lean development, supplementary scripts recomputed the main claims from first principles. These scripts evaluated τ_3^* by exact linear programming, checked the dual fractional packing LP separately as a consistency test, performed the integer optimization in exact rational arithmetic, and used exhaustive enumeration where feasible. The supplied audit report records no discrepancy.

The reported checks include:

- the vertex-copy inequality and its two cancellation identities over all non-isomorphic graphs on up to 7 vertices and all nonadjacent pairs;
- discrete convexity, clone-class progress without splitting, and chordality preservation, including a C_4 test showing why the simpliciality hypothesis is necessary;
- the terminal characterization over all chordal graphs on up to 7 vertices;
- the complete-split formulas of Proposition 6.2 and Corollary 6.3 against the full LP in the tested range;
- the identity $\max_{p+q=n} \Phi_\tau(S_{p,q}) = \lfloor (2n+1)^2/24 \rfloor$ for $1 \leq n \leq 2000$;
- the global maximum over chordal graphs, exhaustively for $n \leq 7$, with additional random tests for $8 \leq n \leq 12$.

The audit scripts, reports, and SHA-256 manifests accompany the preprint. They corroborate the proof and help make the release reproducible; they are not a substitute for the proof.

Scope

The result proved here is the exact finite extremum of the fractional functional Φ_τ. It is not a clique-partition theorem and does not use integral triangle packing. Within the Erdős Problem #81 program, the role of this paper is to identify the complete-split terminal family and the exact fractional value.

Corollary 7.3 (residue-class formula)

Thus the extremal value is one of these two integers according to the residue of n modulo 3. This makes explicit the odd-square-modulo-24 fact used in the case analysis of Proposition 7.1. The same residue class $n \equiv 1 \pmod 3$ is exactly the two-value tie case recorded by Fsat_arg max_tie. The residue fact is formalized as PaperII.odd_sq_emod_24 and the closed form as PaperII.phiTau_max_closed (axiom footprint propext, Classical.choice, Quot.sound).

Lemma 5.1

The accompanying Lean formalization proves this lemma by an equivalent argument that does not use clique trees — via simplicial elimination and the fact that every minimal vertex separator of a chordal graph is a clique — rather than the clique-tree proof given below; see §8. The statement proved is identical.

Theorem 1.1

with p an integer nearest to $(2n+1)/6$. Among complete-split graphs $S_{p,n-p}$, the maximizing integer p is unique unless $n \equiv 1 \pmod 3$, in which case exactly two consecutive values of p attain the maximum; see the case analysis in the proof of Proposition 7.1 and the formalized declarations Fsat_argmax_unique and Fsat_argmax_tie. No uniqueness statement for arbitrary chordal extremizers is claimed.

Corollary 1.2′ (sharpened asymptotic)

After the explicit linear term $n/6$ is extracted, the remaining error is bounded: one has $M(n) = n^2/6+n/6+\theta_n$ with $\theta_n \in (-1,\, 1/24]$. This is immediate from $24\,M(n) \le (2n+1)^2 < 24\,M(n)+24$ and $(2n+1)^2 = 4n^2+4n+1$. Formalized for $n : \mathbb{Z}$, with no lower-bound hypothesis, as PaperII.phiTau_max_sandwich (axiom footprint propext, Classical.choice, Quot.sound).

Remark (coherence with Paper I's bound)

where the right-hand side is the finite estimate $|E(G)| - 2\nu_3^*(G) \le n^2/6 + n/2$ from Paper I [4]. This comparison is numerical rather than a consequence of class containment, since the chordal class is larger than the split class. The slack is $\big(n^2/6+n/2\big) - M(n) = n/3+O(1)$. The inequality is formalized as PaperII.phiTau_max_le_paperI_bound (axiom footprint propext, Quot.sound).

Definition (copy defect)

For nonadjacent u, v set

$$\Delta_{\Phi_\tau}(u,v) = \Phi_\tau(G_{v\to u}) + \Phi_\tau(G_{u\to v}) - 2\Phi_\tau(G) \geq 0 \quad \text{(Lemma 3.1)},$$
$$\Gamma_{\Phi_\tau}(u,v) = \max\{\Phi_\tau(G_{v\to u}), \Phi_\tau(G_{u\to v})\} - \Phi_\tau(G).$$

Then $\Gamma_{\Phi_\tau} \geq \Delta_{\Phi_\tau}/2$, formalized as copyDefect_nonneg and copyGamma_ge_half_copyDefect. The quantity Δ_{Φ_τ} is the resource that any future stability statement must control; it is recorded here as a quantitative handle, not as a stability theorem.

Corollary 7.2 (level sets of the cover functional)

Write $M = 2n+1$. For p in the saturated branch $q \geq p-1$ and every $\delta \geq 0$,

$$\Phi_\tau(S_{p,q}) \;\geq\; \frac{M^2}{24} - \delta \qquad \Longleftrightarrow \qquad \left|p - \frac{M}{6}\right| \;\leq\; \sqrt{\frac{2\delta}{3}},$$

with δ measured from the continuous maximum $M^2/24$. This is an exact equivalence, obtained from identity (7.1) by monotonicity of the square, and is formalized as level_set_iff. In particular Φ_τ is not flat in any direction of p, and its level sets are intervals of radius $\sqrt{2\delta/3}$; within the complete-split family this settles the second half of the stability question raised in Paper III, §12.4.

References

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[2] M. C. Golumbic, *Algorithmic Graph Theory and Perfect Graphs*, second edition, Annals of Discrete Mathematics 57, Elsevier, Amsterdam, 2004.

[3] J. R. S. Blair and B. Peyton, "An introduction to chordal graphs and clique trees," in *Graph Theory and Sparse Matrix Computation*, IMA Volumes in Mathematics and its Applications 56, Springer, New York, 1993, pp. 1–29.

[4] J. P. Traverso Gianini, "Affine profile reduction for fractional triangle packings in split graphs" (Paper I in the series), preprint, July 2026.

[5] L. de Moura and S. Ullrich, "The Lean 4 Theorem Prover and Programming Language," in *Automated Deduction – CADE 28*, Lecture Notes in Computer Science **12699**, Springer, 2021, pp. 625–635, https://doi.org/10.1007/978-3-030-79876-5_37.

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[7] J. P. Traverso Gianini, *Erdős #81 Chordal Clique Partitions: Public Preprints and Formalization Artifacts*, public project repository, 2026, https://github.com/jtraverso/erdos-81-chordal-clique-partitions, accessed July 7, 2026.

[8] T. F. Bloom, "Erdős Problem #81," *Erdős Problems*, https://www.erdosproblems.com/81, accessed July 6, 2026.